

PADER Starter Guide

Simplified Guidance for the GenAR Engineering Challenge

Important

This is a **simplified engineering exercise**, not a complete regulatory PADER authoring guide.

For this exercise, assume you are preparing a **PADER-style report** for one marketed product using the supplied safety dataset (see Appendix B — your specific dataset covers a one-year reporting period).

The real U.S. regulatory basis for PADERs is **21 CFR 314.80(c)(2)**.

The regulation requires, among other things, a narrative summary and analysis, analysis of 15-day Alert reports, a history of safety-related actions, and an index/listing of relevant adverse drug experiences.

You do not need to reproduce every regulatory requirement.

Your goal is to build an AI system that can turn safety data into a **structured, evidence-based report**.

1. Reporting Period

The report should clearly identify:

- Product
- NDA / application identifier if supplied
- Reporting period
- Report type
- Data cut-off / reporting period

Example:

Product: Example COVID-19 Vaccine Report Type: Quarterly PADER Reporting Period: January 1 – March 31, 2026

(The example above uses a quarterly period to illustrate the header format. Your actual reporting period is whatever the supplied dataset covers — see Appendix B.)

2. Narrative Summary and Analysis

This is the main analytical part of the report.

The objective is to provide an overall understanding of the safety information received during the reporting period.

Your system should consider information such as:

- Total number of cases
- Serious vs. non-serious cases
- 15-day Alert cases, if present in the supplied dataset
- Major reported reactions
- Serious reactions
- Patient characteristics
- Outcomes
- Geographic distribution
- Changes or notable patterns
- Other important observations supported by the data

The system should distinguish between:

Observed data

143 cases reported headache.

Derived analysis

Headache was the most frequently reported reaction.

Interpretation

The increase in headache reports may warrant further review.

The AI should **not automatically turn an observation into a medical conclusion.**

3. Summary Analysis of Cases

The system should provide useful aggregate analysis of the cases.

At minimum, consider:

Case volume

- Total cases

- New cases during the reporting period
- Serious cases
- Non-serious cases

Patient characteristics

- Age groups
- Sex
- Country / region

Case characteristics

- Reaction
- Seriousness
- Outcome
- Reporter type, if available
- Product / indication, if available

The exact analyses may depend on the fields provided in the dataset.

4. Reaction / Adverse Event Analysis

Analyze the adverse reactions reported during the period.

Examples:

- Most frequently reported reactions
- Serious reactions
- Reactions by System Organ Class, if MedDRA information is available
- Reactions by age group
- Reactions by sex
- Reactions over time
- Other meaningful patterns

Do not infer information that is not present in the dataset.

5. Serious Cases / 15-Day Alerts

The U.S. regulation specifically requires analysis of 15-day Alert reports submitted during the reporting interval.

For this exercise, if the supplied dataset identifies Alert cases, analyze them separately.

Consider:

- Number of Alert cases

- Seriousness
- Expected / unexpected status if available
- Reaction
- Outcome
- Date received
- Date submitted, if available
- Relevant case identifiers

If detailed narrative information is provided, the system may summarize it.

Do not invent case narratives.

6. Trends and Important Observations

Look for meaningful observations in the reporting data.

Examples:

- Increase or decrease in case volume
- Increase in a particular reaction
- Concentration in a particular population
- Changes by country
- Changes in seriousness
- Changes in outcomes
- Other unusual patterns

However:

A statistical or numerical pattern is not automatically a safety signal.

The system should present an observation and allow a qualified human reviewer to determine its significance.

For example:

"Reports of Reaction X increased from 12 cases in January to 31 cases in March."

is preferable to:

"Reaction X represents a confirmed emerging safety signal."

unless the supplied evidence actually supports such a conclusion.

7. History of Actions

The regulation requires a history of actions taken since the previous report because of adverse drug experiences, such as labeling changes or studies initiated.

For this exercise, this information may be provided separately as structured data or documents.

Examples:

- Labeling change
- Regulatory communication
- Safety-related study initiated
- Additional monitoring
- Risk-minimization action
- Other safety-related action

If no action information is provided:

Do not invent actions.

The system should explicitly indicate that the supplied information contained no reported actions, if that is the defined dataset condition.

8. Case Index / Listing

The report should contain or link to a structured case listing.

At minimum, consider:

- Case ID
- Reaction / adverse event
- Seriousness
- Reporting date
- Country
- Outcome

The exact fields depend on the supplied dataset.

The important principle is:

A reviewer should be able to trace aggregate information back to individual cases.

Official References

- **21 CFR 314.80** — Postmarketing reporting of adverse drug experiences (U.S. PADER basis)
- **FDA — Providing Postmarket Periodic Safety Reports in the ICH E2C(R2) Format** (when PBRER applies instead)
- **EMA GVP Module VII** — Periodic Safety Update Report (EU PSUR framework)
- **ICH E2C(R2)** — Periodic Benefit-Risk Evaluation Report (PBRER structure)

See the main challenge doc for how AI/deterministic split, context assembly, human review, and evaluation are scoped for this exercise — not repeated here.

Appendix A — Key Terms

Term	Meaning
Case / ICSR	One Individual Case Safety Report — one patient, one reporter, one or more reactions, submitted once (possibly updated later).
Reaction	A specific adverse event within a case, coded to a standard vocabulary (MedDRA). A case can have more than one reaction.
Serious	A case is "serious" if it meets ≥ 1 of: caused death , was life-threatening , caused/prolonged hospitalization , caused persistent disability , caused a congenital anomaly , or was medically important for another reason ("other"). Not mutually exclusive — one case can meet several.
Outcome	What happened to the patient for that reaction: recovered/resolved, recovering, not recovered, recovered with sequelae, fatal, unknown.
Expectedness (labelled / unlabelled)	Whether a reaction is already listed in the product's approved label. Determining this requires the product's label as a separate reference — see Appendix B.
Reporter qualification	Who reported it — physician, pharmacist, other health professional, or consumer.
15-day Alert	A serious, expedited case meeting criteria requiring faster reporting to the health authority — see §5.

Appendix B — About Your Dataset

`Bisoprolol_icsr_sample_1068rows.csv` uses a realistic ICSR line-listing structure (E2B/FAERS-style fields) for **Bisoprolol**, a beta-blocker used mainly for hypertension and heart failure. The data is synthetic/derived from publicly available sources, not real patient records — see `DATA_USAGE_NOTICE.md`.

A few things worth knowing before you start parsing it:

- **The reporting period is one year.** The exact start/end dates are in the data itself (`receivedate` field) — nothing else in these guides restates them.
- **1,068 rows $\neq$ 1,068 cases.** There are **1,024 unique `safetyreportid` values** — some cases have more than one reaction row. Decide early how you're deduplicating for case-level counts vs. reaction-level counts, and be consistent about it.

- Almost every case is flagged **serious** (1,023 of 1,024) and meets `fulfillexpeditecriteria` (1,023 of 1,024) — for this dataset, "serious" and "expedited/alert" are nearly the same population. That's normal for spontaneous ICSR data; don't assume something's broken.
- The seriousness columns (`seriousnessdeath` , `seriousnesslifethreatening` , `seriousnesshospitalization` , `seriousnessdisabling` , `seriousnesscongenitalanomaly` , `seriousnessother`) are independent yes / no flags, not mutually exclusive.
- There is **no System Organ Class (SOC) field** — only MedDRA Preferred Term (`patient_reaction_reactionmeddrapt`). Per §4, don't infer SOC groupings that aren't in the data; report at the PT level, or explicitly note SOC-level analysis is unavailable.
- Country appears twice (`occurcountry` vs `primarysource_reportercountry`) — usually the same, occasionally not. Pick one and note which.
- Ages arrive two ways: a coarse `patient_patientagegroup` (mostly blank) and `patient_patientonsetage` — bucket from the numeric field yourself.
- **No "history of actions" data is supplied** with this dataset. Per §7: don't invent any — your report should explicitly state that none were provided for this exercise.
- **No product label/CCDS is supplied**, so expectedness (Appendix A) isn't answerable from this dataset alone. Treat it as out of scope for Version 0.

This guide is a simplified teaching aid for the GenAR AI Engineering Challenge, adapted from the public regulatory references above. It is not a reproduction of any real submitted regulatory report or internal report template.